

ACEA MEMBERSHIP MANAGEMENT SYSTEM (MMS)

SYSTEM ANALYSIS AND DESIGN

1 Introduction

This chapter presents the system analysis and design for the ACEA Membership Management System (MMS). The purpose of the proposed system is to automate the club membership life cycle from application submission, proposer and seconder endorsement, committee screening, payment verification, and member activation to annual subscription tracking, guest control, and reporting. The design has been aligned to the Aero Club of East Africa membership application form and the amended bye-laws so that business rules are embedded directly into the system logic.

2 Analysis of the Existing Process

The current membership process is predominantly document-driven. Applicants fill a paper-based or scanned form, attach supporting documents, obtain proposer and seconder endorsements, and then wait for manual routing to the manager and committee. The same application may be reviewed by multiple parties using physical signatures or unstructured communication channels. This creates delays, poor visibility of application progress, duplicate data entry, risk of lost documents, and difficulty in enforcing time-based or policy-based rules consistently.

The form itself captures rich personal, professional, aviation, family, payment, and endorsement data. However, when this information is managed manually, searching historical applications, checking whether a proposer qualifies, confirming outstanding subscriptions, and generating lists for interviews or approval meetings becomes inefficient. The club therefore requires a centralized digital platform that supports traceability, validation, and reliable reporting.

3 Requirements Analysis

3.1 Functional Requirements

Requirements	Description
Applicant registration and profile capture	System shall capture all ACEA 2025 membership form data including personal details, contacts, company and role, aviation affiliation, emergency contact, spouse and child details, election type, declarations, consent, and other-club membership where declared.
Pre-application visit logging	System shall record the candidate's pre-application club visits, host member, visit date, and notes, and verify that the minimum required visits have been completed before the application proceeds.

Supporting document upload and verification	System shall allow attachment and verification of ID or passport, passport photo, CV, pilot licence where applicable, and any other supporting evidence required during screening.
Eligibility and rule validation	System shall validate mandatory fields, age and date formats, proposer and seconder tenure, membership-type rules, consent capture, document completeness, and configurable club-policy thresholds.
Endorsement management	System shall record proposer and seconder endorsements including member category, years known, years of continuous membership, comments on personal knowledge, integrity and value contribution, signatures, and eligibility verification status.
Interview and committee ballot management	System shall schedule interviews, capture interview notes and panel members, record committee ballot votes, and enforce quorum and adverse-vote rules before an application can advance.
Committee approval and management authorization	System shall capture committee decisions, waitlist or rejection outcomes, and the General Manager and Chairman authorization/signature steps with timestamps and remarks.
Payment and fee schedule management	System shall calculate and record entrance fee and annual subscription charges by membership type, including under-30 fee rules where applicable, and store amount paid, receipt number, payment date, and mode of payment.
Member activation and account provisioning	System shall generate a membership number, create the member record, and optionally provision a system user account only after approvals, required signatures, and payment conditions are satisfied.
Subscription monitoring and member lifecycle control	System shall track yearly renewals, identify members to be posted after 28 February, mark removals after 30 April where unpaid, and retain continuous-membership dates needed for eligibility and discount checks.
Dependent, child and guest management	System shall manage spouse and dependent records, flag children approaching the age threshold for independent membership, and enforce guest limits of six at one time, two visits per month, and twelve visits per year for the same guest.
Reciprocal club and visitor management	System shall maintain approved reciprocal clubs, nominated-club exceptions, reciprocal visitor identities, home-club details,

	membership card or introduction-letter references, visit history, and access-decision records.
Reciprocal visitor search and access decisioning	System shall allow reception, manager, or administrative staff to search reciprocal visitors by name, home club, card number, membership reference, and recorded identity details in order to determine agreement status, remaining usage days, radius-rule compliance, and any block or denial status.
Dashboards, reports, audit trail and notifications	System shall provide dashboards and reports for applications, interviews, approvals, arrears, active members, dependents, guest usage, reciprocal access, payments, and exceptions; log all status changes; and send notifications for missing documents, interview schedules, approvals, and payment reminders.

3.2 Non-Functional Requirements

Quality Attribute	System Expectation
Performance	Application pages should load quickly and common transactions such as saving an application or searching a member should complete in a few seconds under normal club usage.
Usability	Interfaces should be form-driven, clearly labeled, and easy to use by applicants and administrators with varying levels of technical experience.
Reliability	System should preserve submitted data, prevent accidental loss of documents, and maintain transaction consistency between API and database layers.
Security	The solution must implement authentication, authorization, encrypted transport (HTTPS), secure password hashing, audit logs and restricted access to sensitive identity records.
Scalability	The architecture should support future modules such as event bookings, reciprocal club management, and additional digital payment integrations.
Maintainability	Separation of concerns through React frontend, ASP.NET Core backend and Entity Framework data access should simplify testing and long-term maintenance.
Availability	The platform should be accessible during working hours with reliable backup and restore procedures.
Compliance	The design should support data consent capture and responsible handling of personal information in line with the club form statements and applicable data protection obligations.

3.3 Business Rules Derived from Club Policies

The membership management system is rule-driven. Accordingly, the following business logic must be enforced:

- ✓ An applicant must be proposed and seconded by continuing Life, Full, Country or Overseas members with at least three consecutive years of membership.
- ✓ The candidate should have visited the club at least three times before admission; the application must be accompanied by a CV, entrance fee and annual subscription.
- ✓ At least four committee member approvals plus the General Manager authorization are required.
- ✓ Temporary members are restricted from voting, reciprocation privileges and introducing guests.
- ✓ Annual subscriptions fall due on 1 January; unpaid members are posted after 28 February and removed after 30 April.
- ✓ Children are treated as dependents until age twenty-one.
- ✓ Guest access must observe the monthly and annual limits defined in the bye-laws.

4 System Design

4.1 Overall System Architecture

The ACEA MMS adopts a three-tier architecture.

- ✓ The presentation layer contains the React and TypeScript web interface accessed by applicants, administrators, committee members, and the general manager.
- ✓ The application layer contains the ASP.NET Core Web API, where business validation, workflow management, reporting logic, authentication, and audit functions reside.
- ✓ The data layer contains the SQL Server relational database accessed through Entity Framework Core.

This arrangement improves maintainability and supports clean separation of interface, logic, and persistence responsibilities.

ACEA MMS System Architecture

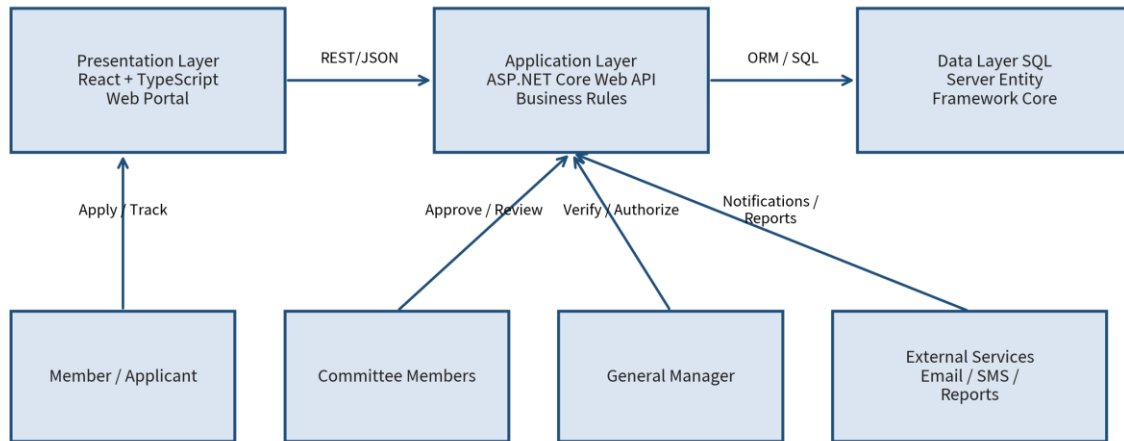


Figure 3.1: ACEA Membership Management System architecture

4.2 Use Case Analysis

The principal actors are the Applicant, Committee Member, General Manager, and Finance/Administrative Staff. The applicant initiates the process by completing the membership form and uploading required documents. Committee members review applications, read proposer and seconder endorsements, and record screening decisions. The general manager confirms final authorization, while finance staff record subscription and entrance-fee payments. The system coordinates these actors, preserves history, and enforces all policy checkpoints before activation.

Use Case Diagram - ACEA Membership Management System

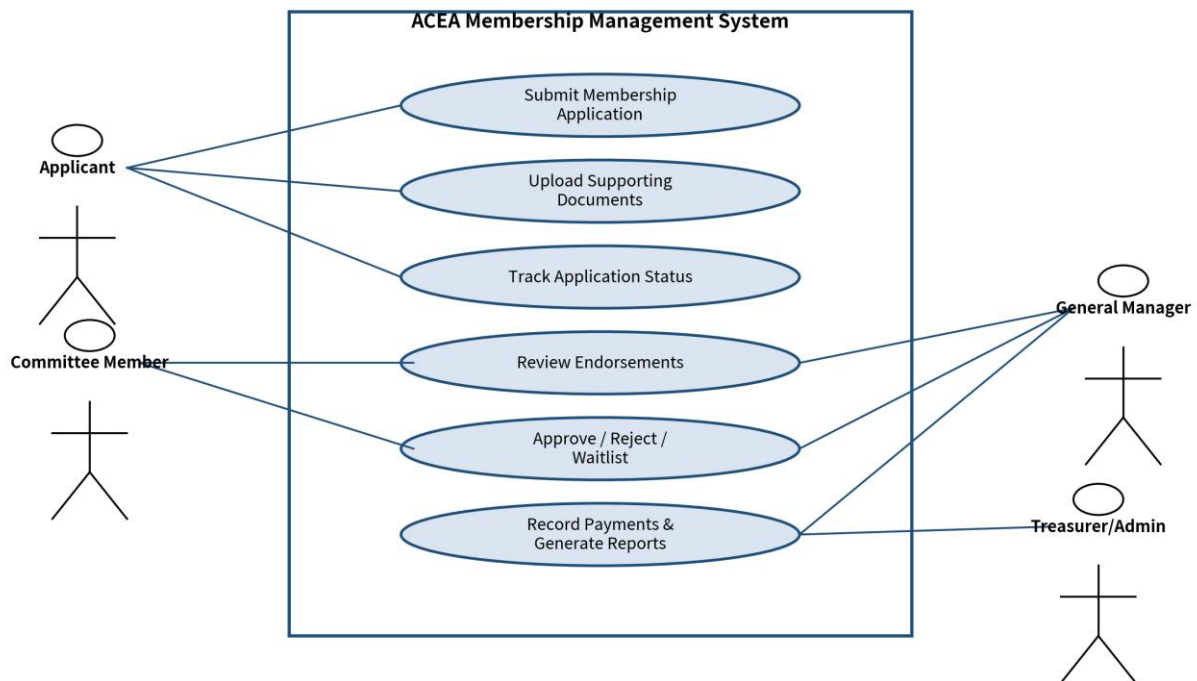


Figure 3.2: Use case diagram for the ACEA Membership Management System

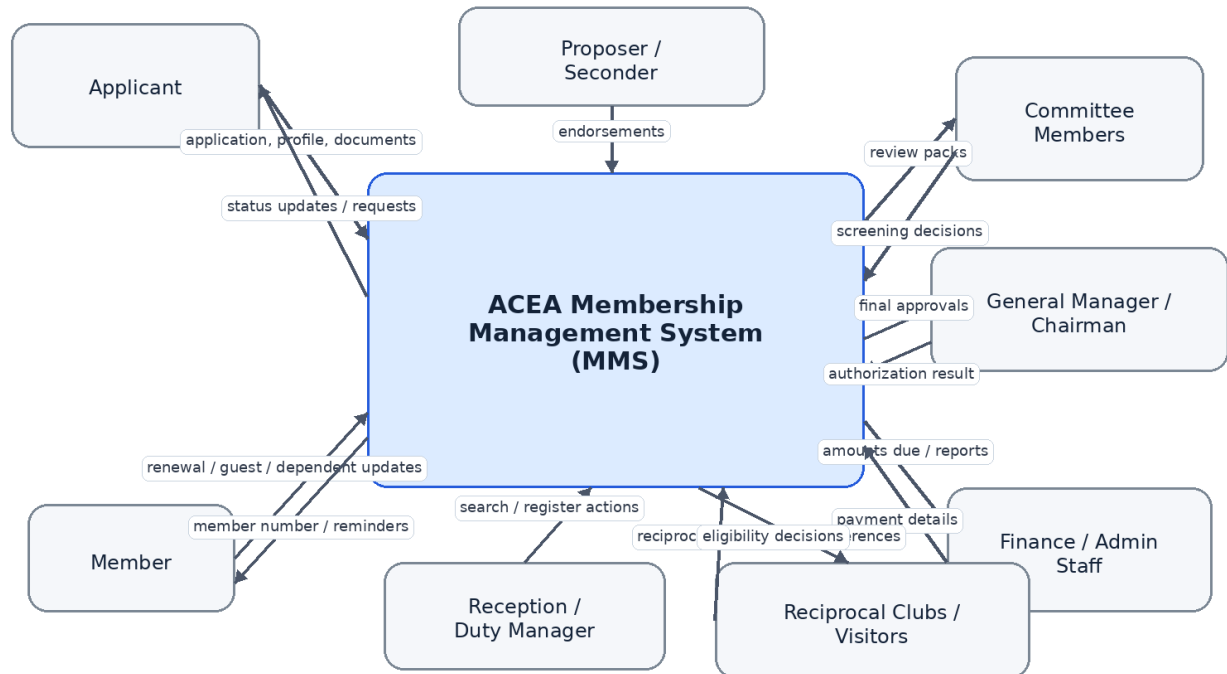
Use Case	Primary Actor	Description
Submit Membership Application	Applicant	Creates new application and enters required membership data.
Upload Supporting Documents	Applicant	Uploads ID/passport, CV, photograph and optional pilot license.
Track Application Status	Applicant	Views whether application is pending, waitlisted, approved or rejected.
Review Endorsements	Committee Member	Checks proposer/seconder details and narrative justification.
Approve / Reject / Waitlist	Committee Member, General Manager	Captures formal decision and remarks in workflow sequence.
Record Payments & Reports	Treasurer/Admin	Registers fees and produces payment and membership reports.

4.3 Data Flow Diagrams

The revised context diagram presents the ACEA MMS as a single system interacting with applicants, proposers and seconders, committee members, the General Manager/Chairman, finance or administrative staff, members, and reciprocal-access staff or visitors. It shows the system receiving applications, endorsements, documents, payments, renewal actions, guest or dependent updates, and reciprocal identity evidence, while returning status updates, approval requests, payment confirmations, membership outcomes, and reciprocal eligibility decisions. The level 1 diagram decomposes the system into profile capture, rule validation, endorsement and committee workflow, member-finance lifecycle management, reciprocal access control, and reporting or audit processes

supported by separate data stores for profiles and rules, applications and workflow, member-finance records, reciprocal records, and audit history.

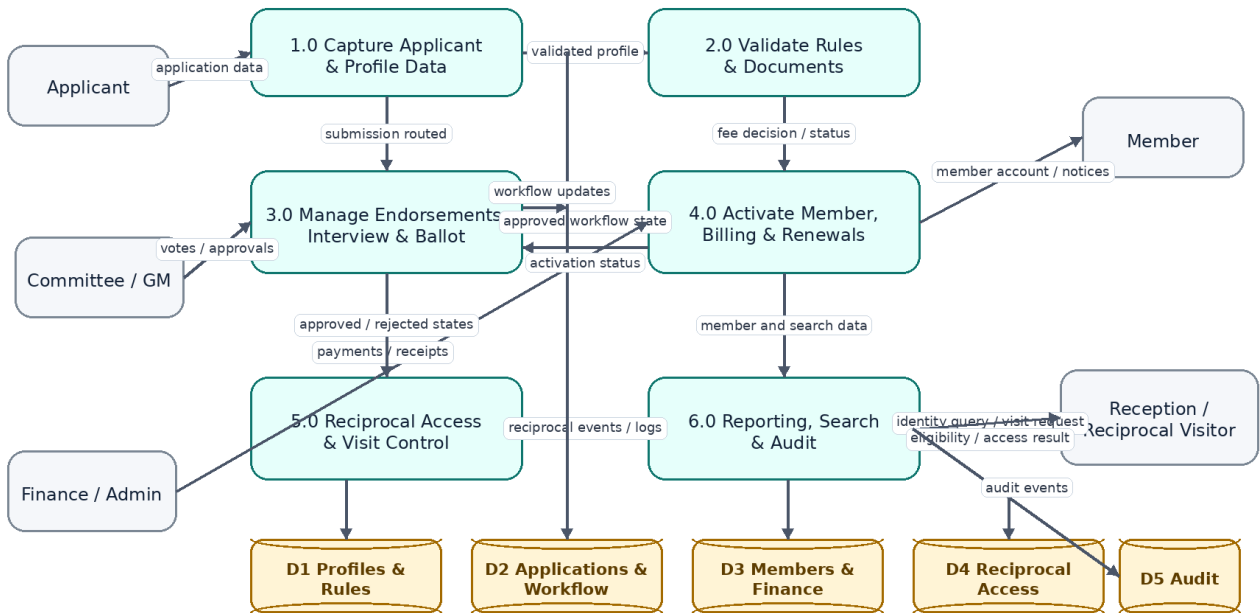
Figure 3.3 Context Data Flow Diagram — ACEA Membership Managemen



External actors send and receive data for application processing, approvals, payments, member lifecycle, and reciprocal access control.

Figure 3.3: Context data flow diagram

Figure 3.4 Level 1 Data Flow Diagram — ACEA Membership Management System



Processes decompose the MMS into capture, validation, approval workflow, member-finance lifecycle, reciprocal control, and reporting/audit services.

Figure 3.4: Level 1 data flow diagram

4.4 Entity Relationship Design

The conceptual database design is organized into six linked domains. The identity and profile domain centres on PersonProfile and its related records : AviationProfile, Companies, ConsentRecords, PreApplicationVisits, and ExternalClubMemberships : which capture applicant identity, aviation affiliation, employer details, consent, required pre-application visits, and other-club membership declarations. The application workflow domain links Applications to MembershipTypeLookup, ApplicationDocuments, Endorsements, Interviews, InterviewPanel, CommitteeBallotVotes, CommitteeApprovals, and ApplicationStatusHistory so that submission, interview, endorsement, ballot, and sign-off activities remain traceable end to end. Once approved, an application creates a Member record in the member-lifecycle domain, which connects to Payments, Subscriptions, Dependents, GuestVisits, and optionally UserAccounts. Governance tables: CommitteeMembers, Officers, and GeneralMeetings : support committee composition, officer terms, quorum checks, and election controls. Reciprocal access is modeled separately through ReciprocalClubs, ReciprocalMembers, ReciprocalVisitRegisters, ReciprocalAccessDecisions, and ReciprocalClubExceptions so agreement status, 30-day rolling 12-month usage, and the 50-kilometre nominated-club exception can be

enforced cleanly. Shared rule and control support is provided by MembershipEligibilityRules, SeniorDiscountRules, MembershipFeeSchedule, ClubSettings, and AuditLogs.

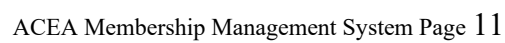


Figure 3.5: Updated entity relationship diagram for the ACEA Membership Management System

Figure 3.6: Simplified class diagram

4.6 Sequence Design

The sequence below illustrates the typical approval process for a new membership application. The applicant completes the form in the web interface, the API persists data and documents, the committee reviews the record, the decision is returned, and the database is updated. Approved applications can subsequently trigger payment verification and member activation.

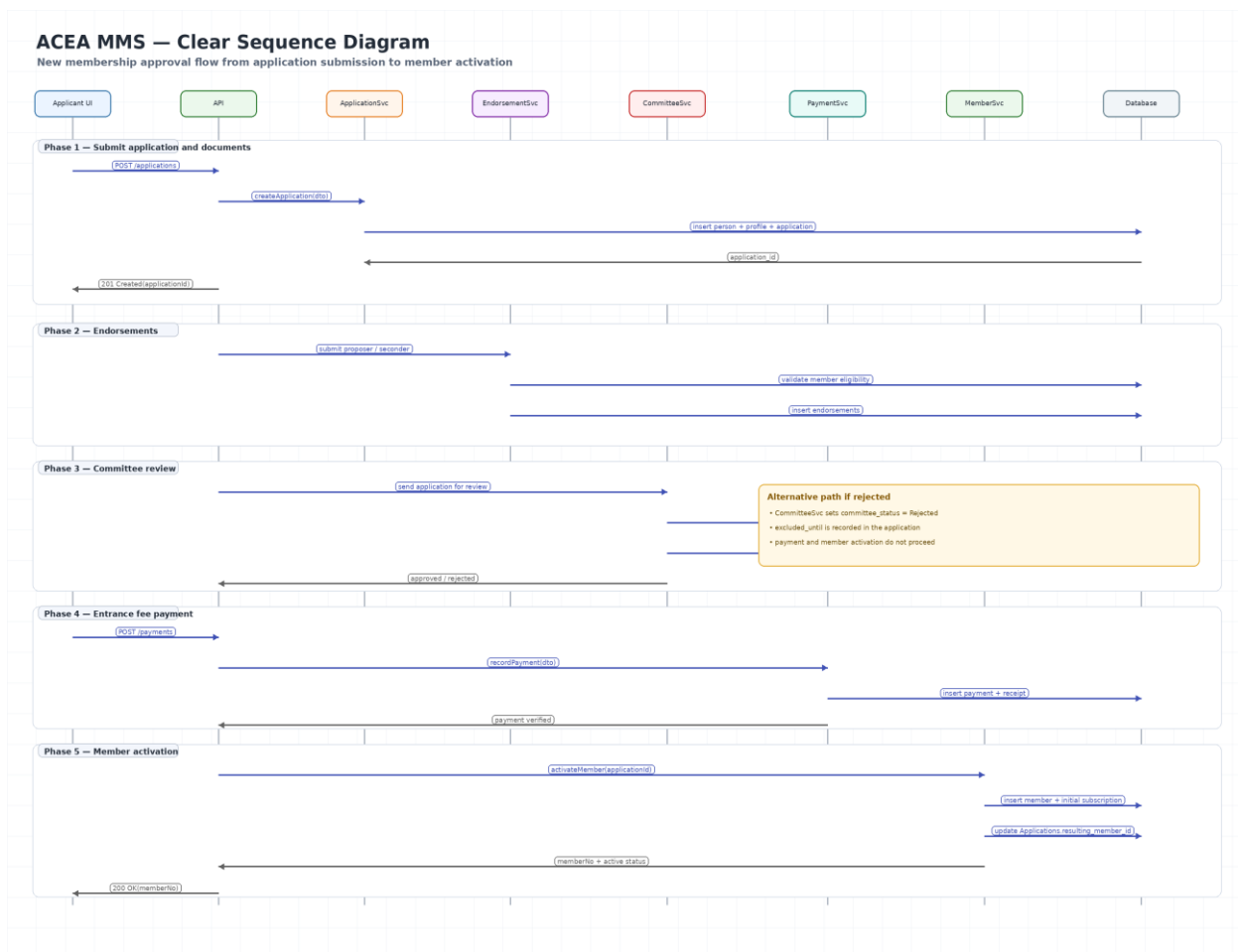


Figure 3.7: Sequence diagram for new membership approval

4.7 Database Schema Design

The database schema translates the conceptual model into normalized relational tables grouped into identity and profile, membership rules, application workflow, governance, finance and member lifecycle, reciprocal

access, and system-control records. Compared with the earlier version, the updated schema explicitly adds pre-application visits, consent capture, external-club membership, interview and ballot records, committee roster and officer-term tables, reciprocal-club exceptions, and an expanded ClubSettings configuration so visit thresholds, signature counts, adverse-vote limits, and quorum values are stored as data rather than hard-coded.

Table	Key Fields	Purpose
PersonProfile	person_id (PK), first_name, middle_name, last_name, id_passport_no, nationality, date_of_birth, gender, blood_group, email, alt_email, mobile_number, tel_other, postal_address, city, state_country, postal_zip_code, country_of_residence, place_of_birth, occupation, company_id (FK), role_at_company, marital_status, spouse_name, spouse_phone, spouse_email, emergency_contact_name, emergency_contact_phone, emergency_contact_email	Master record for every individual the club knows applicant, member, proposer, or seconder created once and reused across applications.
AviationProfile	aviation_profile_id (PK), person_id (FK), holds_pilot_licence, licence_type, licence_number, licence_issuer, owns_or_co_owns_aircraft, aircraft_type, aircraft_registration, hangar_location, aviation_role	Stores the "Affiliation with Aviation" section of the application form (licence and aircraft ownership details).
Companies	company_id (PK), company_name, industry, address, city, country, phone, email	Employer lookup normalizes the Occupation / Company / Role fields off PersonProfile.
ConsentRecords	consent_id (PK), person_id (FK), consent_given, consent_date, withdrawn_date, policy_version	Captures the Declaration & Data Consent tick-box on page 5 of the form and supports personal-data compliance.
PreApplicationVisits	visit_id (PK), person_id (FK), visit_date, member_id (FK, host), purpose, notes, user_id (FK, logged by)	Logs each of the ≥ 3 required club visits so Applications can validate the count before accepting a submission.
ExternalClubMemberships	application_record_id (PK), person_id (FK), club_name, club_scope, still_active, application_id (FK)	Captures the applicant's declared membership in other clubs for screening and profile completeness.
MembershipTypeLookup	membership_type_id (PK), type_code, type_name, has_vote, eligible_for_committee,	Reference table for the 8 membership classes (Full, Country, Overseas, Life, Honorary, Temporary,

	pays_subscription, can_introduce_guests, can_reciprocate	Senior, Senior Life), with their voting / committee / fee rules.
TemporarySubclassLookup	subclass_id (PK), subclass_code, subclass_name, max_days, max_months, fee_applies, requires_guarantor, eligible_for_reciprocation, eligible_to_introduce_guests	Reference table for the three Temporary sub-classes in the Bye-Laws (Temporary, Foreign Air Crew, Special Event).
MembershipEligibilityRules	rule_id (PK), membership_type_id (FK), min_continuous_years, can_propose, can_second, can_vote, can_hold_office, can_introduce_guests, can_reciprocate	Gates who is allowed to propose, second, vote, or hold office, by membership type and tenure, this is what stops an ineligible member from endorsing an applicant.
SeniorDiscountRules	discount_id (PK), membership_type_id (FK), min_age, min_continuous_years, discount_type, discount_value	50% subscription discount at 55+ / 25 years continuous; senior-life privileges at 50 years continuous.
ClubSettings	max_voting_members, max_guests_per_member, reciprocal_max_days_per_12m, reciprocal_radius_km, proposer_seconder_min_years, min_visits_before_application, min_committee_signatures, ballot_adverse_vote_limit, ballot_exclusion_period_years, committee_election_quorum, committee_meeting_quorum, general_meeting_quorum, general_meeting_adjourned_quorum	Single-row configuration table holding numeric limits and workflow thresholds from the Bye-Laws and Articles so policy changes can be applied without schema redesign.
Member	member_id (PK), member_no, person_id (FK), membership_type_id (FK), temporary_subclass_id (FK), status, join_date, continuous_since, is_senior, is_senior_life, guarantor_member_id (FK), chairman_signed_by (FK), elected_date	Stores activated member profiles and renewal status.
Applications	application_id (PK), person_id (FK), membership_type_id (FK), received_date, status, interview_status, manager_status, committee_status, adverse_votes, excluded_until, member_id (FK), entrance_fee_due, annual_sub_due	Stores application lifecycle data.
ApplicationDocuments	document_id (PK), application_id (FK), doc_type, file_name, file_path, verified_flag, uploaded_at	Stores uploaded evidence such as CV and ID.
Endorsements	endorsement_id (PK), application_id (FK), member_id (FK), role, years_known, years_as_member_at_endorsement, eligibility_verified, personal_knowledge_notes, professional_knowledge_notes, value_add_notes, signed_date	Stores proposer and seconder details.

Interviews	interview_id (PK), application_id (FK), interview_date, committee_meeting_ref, outcome, notes	The short interview at a Management Committee meeting required before ballot.
InterviewPanel	panel_id (PK), interview_id (FK), member_id (FK)	Junction table: which committee members sat on a given interview.
CommitteeBallotVotes	ballot_vote_id (PK), application_id (FK), member_id (FK), vote, vote_date	Individual ballot votes cast on a candidate, quorum (7) and adverse-vote exclusion (2) are computed from this table
CommitteeApprovals	approval_id (PK), application_id (FK), member_id (FK), approver_role, decision, remarks, approved_at	Stores the ≥ 4 committee-signature plus General Manager and Chairman sign-off records.
ApplicationStatusHistory	history_id (PK), application_id (FK), status, changed_at, user_id (FK), notes	Timestamped audit trail of every stage transition, visiting, submitted, interview, ballot, approved / deferred / rejected
Officers	officer_id (PK), member_id (FK), office_type — enum(chairman, vice_chairman, treasurer), term_start_date, term_end_date, elected_at_meeting_id (FK), is_current, re_elected	Tracks who holds the three defined Club Offices and their annual retirement / re-election cycle
CommitteeMembers	committee_membership_id (PK), member_id (FK), term_start_date, term_end_date, is_officer, is_actively_engaged_in_aviation, casual_vacancy_fill, elected_at_meeting_id (FK)	Roster of who is currently serving on the Committee and for what term — validates, composition rule (8 Full / Life members, ≥ 6 actively engaged in aviation, plus the Officers).
GeneralMeetings	meeting_id (PK), meeting_type — enum(AGM, EGM), meeting_date, notice_sent_date, notice_days_required, is_special_resolution, quorum_met, attendee_count, chairman_member_id (FK), minutes_recorded, adjourned from meeting_id (FK, self)	The actual AGM/EGM instance that officer elections, ballots, and quorum checks hang off.
MembershipFeeSchedule	fee_schedule_id (PK), membership_type_id (FK), effective_year, joining_fee_standard_kes, joining_fee_under30_kes, annual_subs_kes	Stores the year-versioned joining fee and annual subscription rates by membership type.
Payments	payment_id (PK), application_id (FK), member_id (FK), payment_purpose, amount, receipt_no, paid_on, mode, cheque or mpesa ref, user_id (FK)	Stores entrance fee and subscription payments.
Subscriptions	subscription_id (PK), member_id (FK), subscription_year, amount_due, amount_paid, due_date, posted_date, removed_date	Tracks the annual subscription cycle, including the 28 Feb posting and 30 Apr removal deadlines.
Dependents	dependent_id (PK), member_id (FK), relationship, full_name, date_of_birth, contact_phone, became_own_member_at	Stores spouse and children details.
GuestVisits	visit_id (PK), member_id (FK), guest_name, visit_date, concurrent_guest_seq, month_count, year_count, committee_flagged_undesirable	Supports guest visit limit enforcement (max 6 concurrent, 2/month, 12/year per guest)
ReciprocalClubs	reciprocal_club_id (PK), club_name, club_code, country, agreement_status	Registry of clubs with a reciprocal agreement.

ReciprocalMembers	reciprocal_member_id (PK), reciprocal_club_id (FK), first_name, last_name, membership_card_no, status	Members of reciprocating clubs known to the system.
ReciprocalVisitRegisters	reciprocal_visit_id (PK), reciprocal_member_id (FK), reciprocal_club_id (FK), visit_date, usage_days_last_12_months	Logs reciprocal member visits and running 12-month usage-day totals for enforcing the 30-day cap.
ReciprocalAccessDecisions	decision_id (PK), reciprocal_member_id (FK), decision_type, decision_by_user_id (FK), decision_date	Records access approvals, denials, or restrictions for reciprocal members.
ReciprocalClubExceptions	exception_id (PK), reciprocal_club_name, exempt_from_radius_rule, effective_date, notes	Stores nominated-club exceptions that bypass the 50 kilometre reciprocal-use restriction.
UserAccounts	user_id (PK), member_id (FK), username, password_hash, role, last_login, is_active	Stores authentication data and role assignment.
AuditLogs	audit_id (PK), user_id (FK), action, entity_name, entity_id, details_json, timestamp	Stores traceable system events across every table above.

4.8 User Interface Design Considerations

The interface design emphasizes clarity and guided completion. The public application screen is divided into sections that mirror the official form: personal details, contact information, family information, aviation information, membership election type, emergency contact, proposer/secondor endorsement, attachments, and declaration. Internal users access dashboards showing pending applications, interview-ready cases, missing documents, expiring memberships, and arrears. Search forms support retrieval by name, membership number, category, or status.

Screen	Key Elements
Applicant Form	Step-by-step sections, upload controls, validation messages, declaration checkbox and submit button.
Admin Dashboard	Counts of pending, approved, waitlisted and rejected applications; quick links to screening queues.
Committee Review Page	Applicant summary, proposer/secondor comments, attached files, decision buttons and remarks box.
Finance Page	Amount due, payment mode, receipt capture, renewal history and arrears filter.
Reports Page	Member list, outstanding subscriptions, application funnel, dependent list and export actions.

4.9 Security and Access Control Design

Security is enforced through role-based access control. Applicants can only create and track their own applications. Committee members can review allocated records but cannot alter financial records. Finance staff can manage payments but cannot retroactively change committee decisions. System administrators manage reference data, user roles, and configuration settings. Every update is logged in the audit table to support accountability. Sensitive documents are stored securely and accessed only through authenticated API endpoints.

4.10 Validation Rules

Rule	Validation Logic
Proposer/Seconder tenure	Must belong to approved member classes and have three or more consecutive years of membership.
Pre-application visits	Applicant must have the minimum number of recorded club visits before a valid application can proceed to committee review.
Required attachments	ID/Passport, CV and passport photo are mandatory; pilot licence is conditional on the applicant's aviation profile.
Approval completeness	Application cannot be activated unless interview or ballot rules are satisfied, at least four committee approvals are recorded where required, and management authorization exists.
Payment completeness	Entrance fee and annual subscription must be recorded before membership number generation.
Guest compliance	System must block entries exceeding six guests at a time, two visits per month, or twelve visits per year for the same guest.
Dependent age rule	Children reaching age twenty-one should be flagged for independent membership transition.
Reciprocal access compliance	System must allow reciprocal access only for approved reciprocal clubs, valid card or introduction evidence, successful register entry, compliance with the 30-day rolling 12-month limit, satisfaction of the 50-kilometre rule except for nominated clubs, and confirmation that the visitor is not blocked.